

Bridge-AI Smart mushroom Project progress report

Project Overview

The Bridge-AI project is a key component of JHUB Africa's Smart Mushroom Farming initiative, an IoT-enabled agricultural innovation that leverages AI and sensor technology to optimize mushroom cultivation within modified shipping containers.

Bridge-AI Website Technology Stack

- **Backend Framework:** Python Flask – chosen for its lightweight architecture, flexibility, and ease of deployment
- **Data Storage:** JSON – utilized for lightweight, human-readable data serialization
- **Version Control:** GitHub – enabling collaborative development and change tracking
- **Deployment Platform:** Render – cloud hosting for testing and production environments

Key Development Milestones

Requirements Gathering (complete)

- Participated in stakeholder sessions with project manager and IoT engineers
- Documented user stories for smallholder farmers as primary users
- Defined functional and non-functional requirements for the Bridge-AI website

Backend Development & Frontend Integration (complete)

- Set up Flask application with core routing and request handling
- Implemented JSON-based data models for project information and configurations
- Established GitHub repository with version control workflows
- Developed responsive user interfaces with authentication and dashboard functionality

Website Testing & Deployment (Partial complete)

- Conducted unit, integration, and user acceptance testing
- Optimized application performance and security measures
- Deployed on Render with environment configuration and dependency management
- Prepared deployment documentation and user guides

Farmers' Smart Dashboard Planning (In progress)

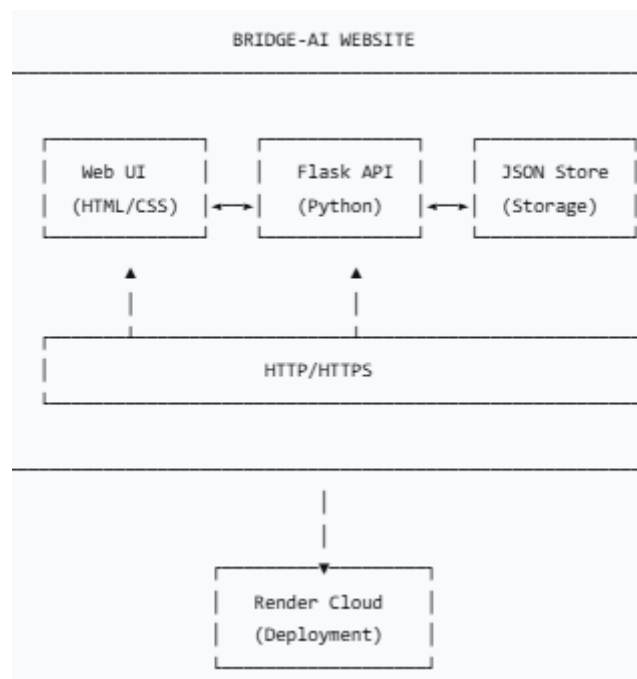
- Began planning the dashboard data model and architecture
- Identified key entities: sensor data, user profiles, grow room configurations, alert settings
- Created initial design concepts for mobile-responsive UI
- Tech stack for dashboard and database to be determined based on project requirements.

Bridge-AI API Architecture

The system follows a REST API-first approach, allowing IoT devices to communicate with the Flask backend using HTTP requests and responses. This architecture supports:

- Real-time data processing and visualization
- Model versioning and tracking
- Scalability for future expansion

System Context Diagram



Key Achievements

1. Successfully developed and deployed Bridge-AI website with Python Flask
2. Implemented JSON-based data storage for project information
3. Established GitHub version control for 16-member team collaboration
4. Deployed on Render for stakeholder testing and demonstration
5. Began planning farmers' smart dashboard with mobile-responsive design

Smart mushroom IOT

Progress Summary

Research phase completed through literature review, environmental requirements analysis, sensor and actuator selection, and IoT architecture definition. The control system was successfully simulated on Wokwi using ESP32 microcontroller with DHT22 sensor, LCD display, and relay controlled actuators including fan, heater, humidifier and lighting. Automatic regulation based on mushroom fruiting stage requirements was validated. The project now advances to structural design phase with mushroom growing room modelling in SketchUp. The successful simulation confirms that the proposed control strategy is feasible before physical hardware implementation.

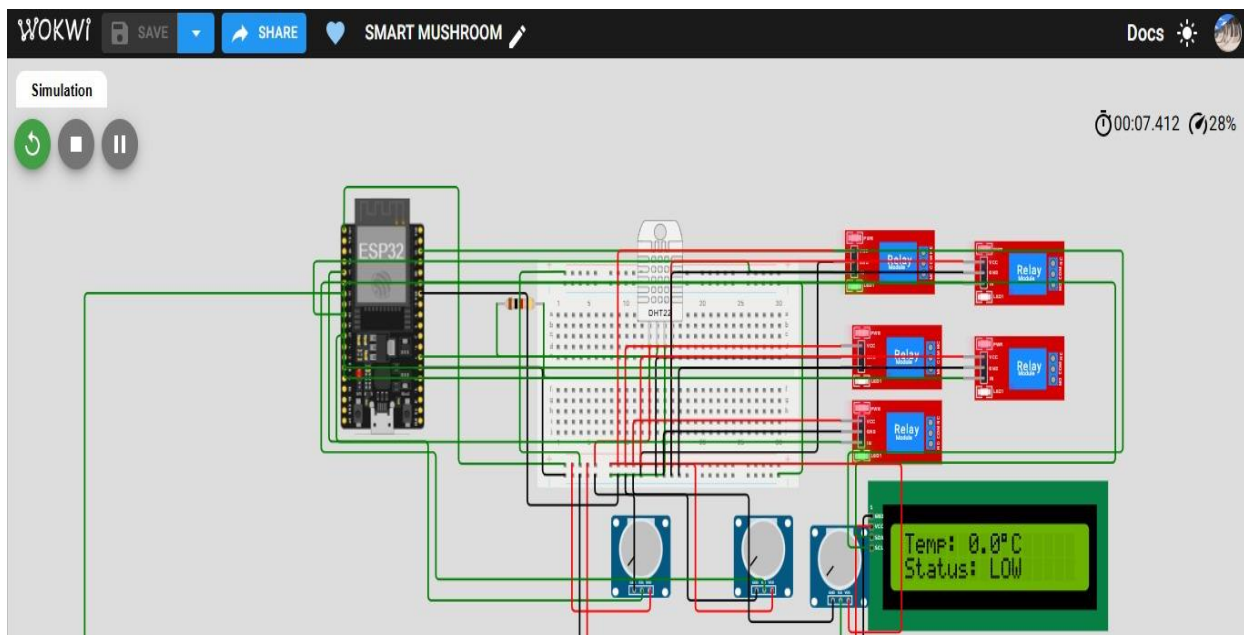


Figure 1. ESP32-based Wokwi simulation showing environmental monitoring and control.

SYSTEM ARCHITECTURE

The IoT subsystem operates within a 20-foot modified shipping container, equipped with sensors and actuators controlled by a central microcontroller (ESP32/Raspberry Pi). The system enables remote monitoring via a smartphone web application, empowering smallholder farmers with precision agriculture capabilities.

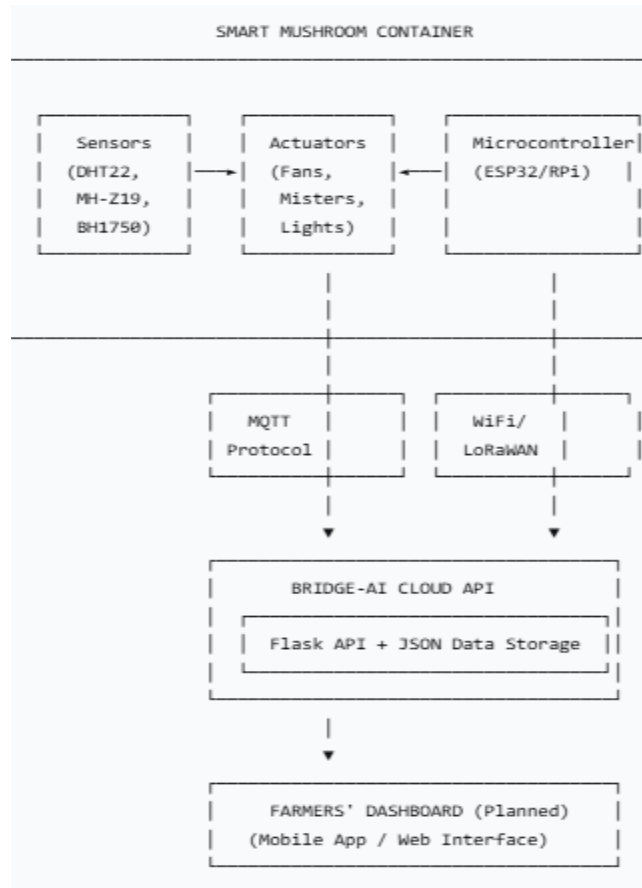
Sensor Components

Sensor	Purpose	Specifications
DHT22	Temperature & Humidity Monitoring	±0.5°C accuracy, 0-100% RH range
MH-Z19	CO ₂ Concentration Monitoring	0-5000 ppm, ±50 ppm accuracy
BH1750	Light Intensity Detection	1-65,535 lux range
Soil Moisture	Substrate Moisture Monitoring	Capacitive sensing

Actuators

Actuator	Purpose	Control Mechanism
Exhaust Fans	Temperature & CO ₂ Regulation	Activated at thresholds
Misting System	Humidity Regulation	Automated misting cycles
UV/LED Lights	Light Cycle Management	Programmable schedules

Data Flow Architecture



Automated Microclimate Control

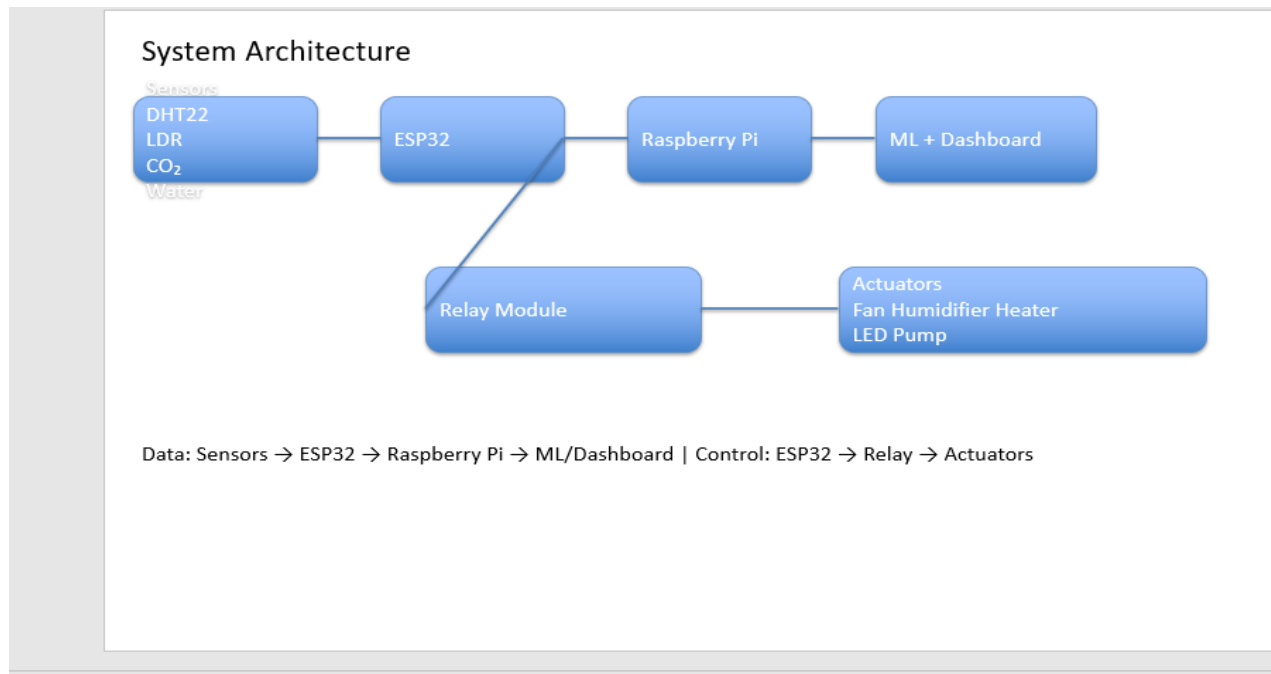
The system implements stage-aware climate control, adjusting environmental parameters based on mushroom growth phases:

- **Temperature:** Maintained at optimal range for each growth stage
- **Humidity:** Controlled between 90-100% for ideal conditions
- **CO₂ Regulation:** Automatic ventilation when thresholds exceeded
- **Light Cycles:** Programmable illumination schedules

Integration with Bridge-AI

- Sensor data streams to API via HTTPs
- Actuator commands processed through AI-driven decision engine
- Real-time alerts and notifications to farmers via smartphone

- Historical data stored for analytics and predictive modeling



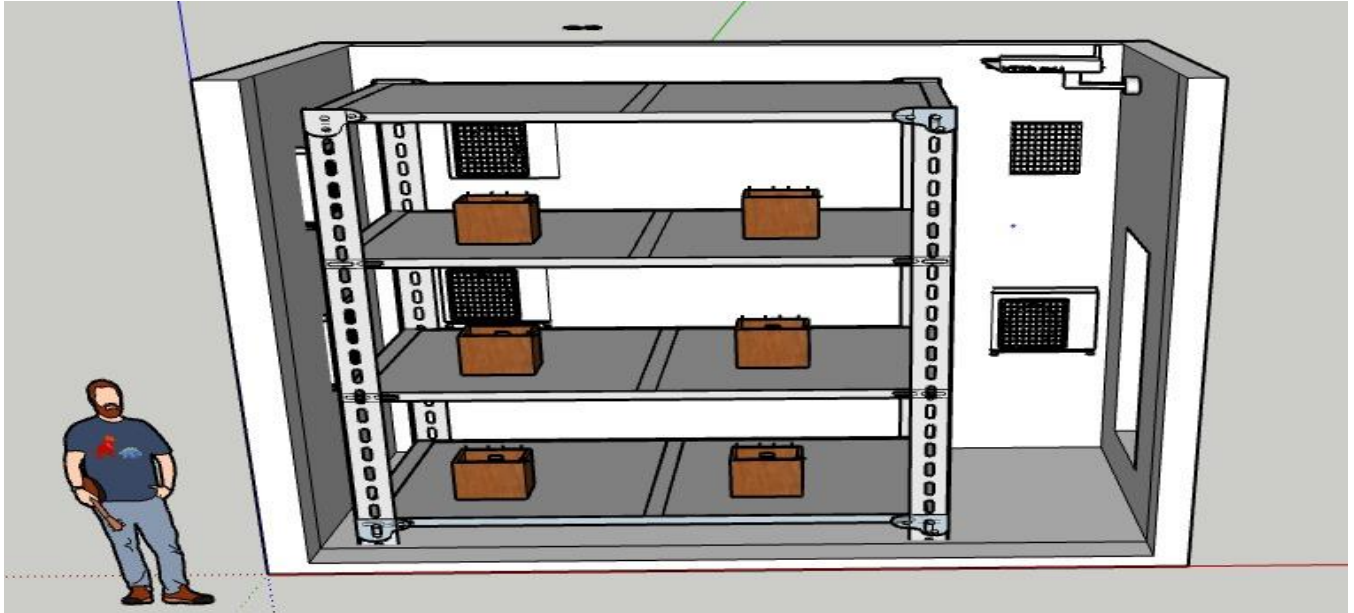
Structural Design Progress

The team is currently finalizing the structural design of the mushroom growing room using SketchUp. The proposed room measures approximately 6 m × 2.4 m × 2.4 m and is designed to provide a controlled indoor environment.

The design includes:

- Insulated walls and roofs.
- Concrete floor.
- Four mushroom growing racks.
- Fresh-air intake and exhaust ventilation.
- Water storage tank.
- Control cabinet housing the ESP32, Raspberry Pi and relay modules.
- Dedicated locations for sensors and actuators.
- Central maintenance walkway.

The room layout has been planned to maximize airflow uniformity, simplify maintenance and reduce overall energy consumption



Caption: Figure 2. Proposed SketchUp mushroom growing room structure.

Current Status

At this stage, the project has successfully completed:

- Research and requirement analysis.
- Selection of hardware components.
- IoT system architecture.
- Virtual simulation and validation.
- Preliminary room structural design.